

## Assignment 1

Due 29/04/2026

### Assignment Title: Designing a Smart Campus Energy Monitoring and Intervention System

#### Scenario

Your university has tasked you with designing the computational architecture for a Smart Campus Energy Monitoring and Intervention System (SCEMIS). The system must:

- Collect real-time energy usage data from thousands of sensors across campus buildings.
- Identify patterns of wasteful energy consumption.
- Predict future energy demand under different weather and occupancy scenarios.
- Recommend targeted interventions (e.g., adjust HVAC set-points, schedule equipment usage) to reduce energy waste.
- Provide a causal analysis of *why* certain interventions work or fail.

You are not required to implement the system; you must design it at the architectural and algorithmic level.

#### General Instructions

- This is an individual assignment.
- Your submission should be a written report with diagrams, pseudocode, and justification for every design decision.
- You must explicitly reference concepts from the course (hierarchical decomposition, functional decomposition, abstraction layers, models/representations, pattern recognition, predictive reasoning, machine learning, statistical thinking).
- Answers that rely on generic, non-specific statements (e.g., “use AI to find patterns”) without concrete technical detail will receive low marks.

### Part A: Problem Decomposition and Abstraction (40 marks)

#### Question 1: Hierarchical Problem Breakdown (15 marks)

Produce a hierarchical decomposition of the SCEMIS into at least four levels, from the overall system goal down to atomic tasks.

- (a) Draw a tree diagram showing at least 15 nodes across the hierarchy. (5 marks)
- (b) For two of the intermediate nodes, explain why further decomposition is useful and what criteria you used to decide when to stop decomposing. (5 marks)
- (c) The system must run on a campus network with limited bandwidth and intermittent connectivity. Explain how your hierarchical breakdown supports graceful degradation. i.e., the system continues to provide useful output even when some lower-level components fail or are disconnected. Provide one concrete scenario. (5 marks)

#### Question 2: Functional Decomposition (10 marks)

Choose the “Intervention Recommendation” sub-system from your hierarchy.

- (a) Perform a functional decomposition of this sub-system. List at least five distinct functions, specifying for each: the function’s name, its inputs, its outputs, and a one-sentence description of what it computes. Present your answer as a table. (5 marks)
- (b) Two of your functions will need to call each other. Explain how you would manage their coupling to ensure that changes to one function do not break the other. Use the terms *cohesion*, *coupling*, and *interface* in your answer. (5 marks)

#### Question 3: Abstraction Layers (10 marks)

Design the software architecture of the sensor-data pipeline as a stack of four abstraction layers.

- (a) Name each layer, describe its responsibility, and list one concrete technology or protocol that could implement it. Draw the stack as a diagram. (5 marks)
- (b) Suppose the university decides to replace all temperature sensors with a different manufacturer’s model that uses a different communication protocol. Explain, using the language of *information hiding* and *interface stability*, which layer(s) would need to change and which would remain unaffected. (5 marks)

#### Question 4: Models and Representations (5 marks)

The SCEMIS needs a model of a building’s thermal behaviour to predict energy demand.

- (a) Propose two different models for a building’s thermal behaviour (e.g., a physics-based model, a data-driven model, a state-machine model, a graph model). Describe what each model captures and what it deliberately ignores. (3 marks)
- (b) Discuss a scenario in which one model would be preferable to the other, justifying your choice in terms of the trade-offs you identified. (2 marks)

### Part B: Pattern Recognition and Data-Driven Reasoning (40 marks)

### Question 5: Identifying Patterns in Large Datasets (10 marks)

The campus has 5,000 sensors, each reporting energy consumption every 15 minutes, generating approximately 480,000 data points per day.

(a) Describe two distinct pattern-recognition techniques (e.g., clustering, sequence mining, anomaly detection, frequent pattern mining) that you would apply to this data. For each technique, state: (i) what kind of pattern it would discover, (ii) what input features the algorithm would use, and (iii) one actionable insight the discovered pattern could provide to the campus energy manager. (6 marks)

(b) The data stream is noisy: sensors occasionally report impossible values (e.g., negative power consumption) or drop out entirely for hours. Explain how your pattern-recognition pipeline would handle such data quality issues before pattern mining takes place. Be specific about the preprocessing steps. (6 marks)

### Question 6: Predictive Reasoning (10 marks)

The SCEMIS must predict hourly energy demand for the next 24 hours.

(a) Propose a predictive model (you may choose any machine-learning or statistical model you have studied). Justify your choice in terms of (i) the nature of the data (time-series, multiple exogenous variables), (ii) the need for interpretability vs. accuracy, and (iii) the requirement to update predictions as new data arrives. (6 marks)

(b) The energy manager observes that on cold days, energy demand is high, and on warm days, energy demand is also high (due to air conditioning). This creates a U-shaped relationship between outdoor temperature and energy demand. A simple linear regression would fail to capture this pattern. Explain, with a diagram or equation, how you would modify your model to handle such non-linear relationships. (6 marks)

### Question 7: Machine Learning as Pattern Recognition (10 marks)

(a) “Machine learning is nothing more than automated pattern recognition.” Critically evaluate this statement with reference to the SCEMIS. In your answer, discuss: (i) one task for which pure pattern recognition is sufficient, (ii) one task for which pattern recognition alone is insufficient, and (iii) what additional capability is needed for the second task. (6 marks)

## Part C: Synthesis and Critical Reflection (20 marks)

### Question 8: Integration (10 marks)

Write a critical reflection that addresses the following:

(a) How did the hierarchical decomposition you designed in Part A influence the pattern-recognition tasks you defined in Part B? Provide two concrete links between a

specific node in your decomposition tree and a specific pattern-recognition or prediction task. (4 marks)

(b) Discuss one ethical concern that arises from deploying an automated intervention system like SCEMIS. Consider issues such as fairness (are some buildings or occupants disproportionately affected?), privacy (what data is collected and who can access it?), or autonomy (do building occupants have the right to override automated decisions?). Propose a design or policy measure to mitigate the concern you identify. (3 marks)

(c) Reflect on what this assignment has taught you about the relationship between problem decomposition and data-driven reasoning. Has designing the system helped you understand either concept more deeply? Be specific. (3 marks)